

Tanvir R. Patwary

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EDUCATION

Georgia State University

B.S. in Computer Science

Atlanta, Georgia

Expected July 2027

- Concentration: Distributed Systems & Software Design | Dean's List: Spring 2026
- Relevant Coursework: Data Structures & Algorithms, System Design, Discrete Math, Linear Algebra, OOP

PROFESSIONAL EXPERIENCE

ADP — Global Product & Technology

Application Developer Intern

Alpharetta, Georgia

May 2026 – Jul 2026

- Built and shipped an MVP AI agent embedded in ADP Workforce Now for the Human Resource Outsourcing (HRO) team, enabling payroll administrators to query employee and payroll data through natural language and reducing manual navigation and search time
- Engineered a hub-and-spokes agent architecture using AWS Bedrock AgentCore, ADP AI Studio, and the Strands Agents SDK, integrating live Workforce Now Payroll APIs through MCP (Model Context Protocol) tools for real-time data retrieval
- Applied Retrieval-Augmented Generation (RAG) to ground the LLM in verified payroll data, ensuring accuracy and preventing hallucination in a financial context
- Optimized LLM latency and prompt engineering, and developed reusable UI components in Storybook to improve agent performance and user experience
- Operated within a full enterprise SDLC using Jira, Git, Bitbucket, Jenkins (CI/CD), and Splunk for distributed service debugging across ADP systems

Georgia Environmental Protection Division

Software Development Intern

Atlanta, Georgia

May 2024 – Jan 2025

- Developed and maintained .NET applications using C# and JavaScript frameworks, resolving 20+ weekly software issues
- Implemented comprehensive unit testing with the NUnit framework, executing 100+ tests and increasing application reliability by 25%
- Deployed applications to AWS cloud infrastructure maintaining 95% uptime, and collaborated with an 8-member cross-functional team using Git

PROJECTS

FlowStaff

Personal Project

May 2026 - Present

- Built a full Spring Boot REST API with PostgreSQL, following a layered architecture (Controller → Service → Repository) with DTO-based data protection
- Implemented JWT authentication, BCrypt password hashing, and role-based access control (RBAC) using Spring Security
- Applied AppSec best practices including input validation, mass-assignment protection, and OWASP-aligned data exposure controls

Coffee Shop Application

Personal Project

Nov 2024 – Present

- Developed a Node.js/Express backend with MySQL via Sequelize, integrating Stripe payments and JWT authentication
- Implemented role-based access control and achieved 95% code coverage using the Vitest testing framework

ACTIVITIES AND LEADERSHIP

Georgia State University Muslim Student Association

Treasurer

Atlanta, Georgia

Current

- Manage the organization's budget and event logistics for 150+ active members, coordinating with campus vendors on events up to \$5,000

SKILLS

Languages: Java, JavaScript, Python, C#, .NET, SQL, HTML/CSS

Frameworks & Tools: Spring Boot, Spring Security, Node.js/Express, REST APIs, JPA/Hibernate

AI & Cloud: AWS Bedrock AgentCore, MCP Tools, RAG, Prompt Engineering, AWS

Security: Application Security, JWT, RBAC, BCrypt, OWASP, Splunk

DevOps & Testing: Git, Bitbucket, Jenkins, CI/CD, Jira, Unit Testing (NUnit, Vitest)

Databases: PostgreSQL, MySQL, Microsoft SQL Server